

# Alvin Le

(408) 807-6328 | [alvinle44@gmail.com](mailto:alvinle44@gmail.com) | [github.com/alvinle44](https://github.com/alvinle44) | San Jose, CA

## SECURITY CLEARANCE

---

**Interim Secret Clearance (Active)**

## EDUCATION

---

### Georgia Institute of Technology

**Atlanta, GA**

M.S. in Computer Science; Specialization in Machine Learning; GPA: 4.00/4.00

Expected December 2026

- **Relevant Coursework:** Machine Learning, Deep Learning, NLP, Agentic AI, Computer Networks, OS, Software Design, Data Structures & Algorithms, AI Ethics, Knowledge Based AI

### University of California, Irvine

**Irvine, CA**

B.S. in Biological Sciences; Dean's Honor List (all quarters); GPA: 3.92/4.00

Sept. 2019 – June 2023

## TECHNICAL SKILLS

---

**Languages:** Python, C/C++, JavaScript/TypeScript, Java, SystemVerilog, x86 Assembly

**Frameworks & Tools:** React, React Native, Next.js, Node.js, FastAPI, Supabase, gRPC, REST APIs, Git

**ML / AI:** PyTorch, scikit-learn, XGBoost, OpenCV, YOLOv8, Transformers, CNNs, GNNs, LLMs, RAG pipelines, LangChain, ChromaDB, Prompt Engineering, Tool-Calling Agents, NumPy, pandas

**Certifications:** NVIDIA DLI — Deep Learning, Diffusion Models, Anomaly Detection, RAG Agents with LLMs, Agentic AI Applications, Rapid LLM Development

## PROFESSIONAL EXPERIENCE

---

### STEM Intern

**Summer 2026**

General Infomatics Inc. (TRMCi Program)

Point Mugu, CA

- DoD-affiliated TRMCi Summer 2026 Internship Program supporting software test & evaluation efforts for defense systems.
- Will perform software testing and evaluation with AI/ML techniques, validating system behavior and contributing to quality assurance.

### Biomedical Research Intern

**Aug. 2022 – Jun. 2023**

UC Irvine — Alzheimer's Disease Lab

Irvine, CA

- Automated ImageJ (Jython) image analysis scripts, improving pipeline throughput by 60%.
- Analyzed 1,000+ microscopy images across a 4-person team; designed experiments and presented quantitative findings on protein PGC1- $\alpha$  in transgenic mouse neurons.

## PROJECTS

---

### AI Shot Tracking App | Python, FastAPI, React Native, TypeScript, YOLOv8, Roboflow

- Built a full-stack mobile app with a React Native/TypeScript frontend and FastAPI backend, exposing APIs for real-time YOLOv8 inference on streamed video.
- Trained a custom YOLOv8 model achieving 94% ball and 92% rim detection accuracy.
- Engineered trajectory classification using polynomial regression and motion heuristics, achieving 80% shot outcome accuracy.

### ECA Datasheet RAG Assistant | Python, FastAPI, LangChain, ChromaDB, OpenAI API

- Building a retrieval-augmented generation (RAG) assistant capable of answering technical questions over hardware component datasheets using semantic retrieval and vector embeddings.
- Engineering PDF text extraction, chunking, embedding generation, and citation-aware retrieval workflows for technical documentation.
- Integrating LangChain retrieval pipelines with LLM APIs to generate grounded responses using retrieved datasheet context and page-level references.

### Distributed File System | C++, gRPC, Protocol Buffers

- Designed and implemented a client-server file system with gRPC APIs supporting file upload/download, metadata synchronization, and consistency validation across concurrent clients.
- Engineered async write locks and caching mechanisms for fault tolerance; implemented conflict resolution for concurrent writes.

### NBA Player Prop Prediction System | Python, XGBoost, scikit-learn, NBA API, The Odds API

- Built an XGBoost-based prediction system trained on 160K+ samples using NBA statistics, historical odds, and injury data.
- Achieved 54% test accuracy; identified and fixed data leakage via temporal filtering and feature corrections.
- Engineered ML pipeline with 50+ features including teammate impact, usage patterns, and automated ROI tracking.

### Movie Recommendation System | Python, PyTorch, LightGCN, GNN

- Developed a graph-based recommendation system on MovieLens-32M with 200K users and 55K movies.
- Implemented BPR loss with negative sampling and optimized graph propagation, achieving 4.5 $\times$  improvement over baseline.